Practical 2e: Using Delegates and Events

AIM: To demonstrate the use of delegates and events in C# for achieving callback mechanisms and event-driven programming.

PROGRAM CODE

```
using System;

namespace DelegatesEvents
{
   // Declare a delegate
   public delegate void NotifyHandler(string message);

   class EventPublisher
   {
      // Declare an event using the delegate
      public event NotifyHandler OnDataProcessed;

      public void ProcessData(string data)
      {
         Console.WriteLine($"Processing data: {data}");
         // Raise the event
         OnDataProcessed?.Invoke($"Data '{data}' has been processed!");
      }
   }

   class EventSubscriber
   {
      public void HandleEvent(string message)
      {
         Console.WriteLine($"[Subscriber] Received: {message}");
      }
   }

   // Simple delegate demo
   delegate int MathOperation(int x, int y);

   class Program
   {
```

```
    static int Multiply(int a, int b) => a * b;
    static int Add(int a, int b) => a + b;

    static void Main(string[] args)
    {
        Console.WriteLine("=== Delegates Demo ===");
        MathOperation op = Multiply;
        Console.WriteLine($"Multiply(4,5) = {op(4, 5)}");

        op = Add;
        Console.WriteLine($"Add(4,5) = {op(4, 5)}");

        Console.WriteLine("\n=== Events Demo ===");
        EventPublisher publisher = new EventPublisher();
        EventSubscriber subscriber = new EventSubscriber();

        // Subscribe to the event
        publisher.OnDataProcessed += subscriber.HandleEvent;
        publisher.OnDataProcessed += (msg) => Console.WriteLine($"[Lambda] {msg}");

        publisher.ProcessData("Student Record");
        publisher.ProcessData("Salary Data");

        Console.ReadKey();
    }
  }
}
```

OUTPUT

```
=== Delegates Demo ===
Multiply(4,5) = 20
Add(4,5) = 9

=== Events Demo ===
Processing data: Student Record
[Subscriber] Received: Data 'Student Record' has been processed!
[Lambda] Data 'Student Record' has been processed!
Processing data: Salary Data
[Subscriber] Received: Data 'Salary Data' has been processed!
[Lambda] Data 'Salary Data' has been processed!
```